

Award Abstract # 1722382**SBIR Phase I: Virtual and Augmented Reality Enabled Personalized Manufacturing Training**

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Rajesh Mehta rmehta@nsf.gov (703)292-2174 TI Translational Impacts TIP Directorate for Technology, Innovation, and Partnerships	Feedback
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History of Investigator:
Rajesh K Jha (Principal Investigator) rkjha1@siminsights.com
Recipient Sponsored Research Office:
SimInsights Inc 93 LUPARI IRVINE CA US 92618-0116 (714)651-4104
Sponsor Congressional District:
47
Primary Place of Performance:
SimInsights 20381 lake forest drive, B15 Lake Forest CA US 92630-8109
Primary Place of Performance Congressional District:
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ABSTRACT

This SBIR Phase I project will research the feasibility of applying virtual reality (VR) and augmented reality (AR) technologies in manufacturing and material testing training programs to increase academic achievement, accelerate progress toward degree and foster more effective use of expensive lab-equipment. According to 2014 statistics, in the United States, less than 40% of students entering college in STEM fields finish with a STEM degree. And even for those who finish, a majority (over 60%) take several years longer than the nominal time of 4 years. Providing students with VR and AR enabled personalized manufacturing labs is a promising way to increase students' engagement and better prepare them for future careers as well as broaden students' outlook on innovation. As described in the recent US council report on competitiveness, advanced manufacturing has the largest multiplier effect on jobs. In order for the US to gain competitiveness in this sector, there is a pressing need for a trained workforce that is able to develop new materials testing protocols and manufacturing processes. This SBIR project aims to develop technologies that enable the education sector to leverage VR and AR to accelerate training.

Proposed software will be evaluated through pilot testing. Technology and insights resulting from this project may find applications in other domains such as life sciences.

Key technical innovations in this proposed research include the development of a novel high-fidelity modular virtual object model, customized task models based on evidence centered design, and methods to flexibly combine object models with task models to rapidly develop activities suitable for virtual reality (VR) and augmented reality (AR) devices. These innovations may significantly reduce the content creation cost and potentially dissolve the boundary between classroom education and hands-on training, providing an effective and seamless educational experience. In the long term, these innovations may also be applied broadly to training in other domains. The overarching goal of this SBIR project is to evaluate the feasibility of the proposed system for content creation and learning analytics and apply them to create innovative content for use in manufacturing courses. The proposed system and content will be evaluated via pilot testing with engineering students to collect rich datasets comprising videos, sensor time series data and simulation-event logs. Data analysis will seek evidence for changes in both students outcomes (changes in content knowledge, engagement, attitude towards engineering, self-efficacy towards learning engineering, usability of the product) and instructor outcomes (attitudes about the usefulness of the product).

PROJECT OUTCOMES REPORT

Disclaimer

This Project Outcomes Report for the General Public is displayed verbatim as submitted the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed in this Report are those of the PI and do not necessarily reflect the views of the National Science Foundation; NSF has not approved or endorsed content.

In this SBIR Phase I project, we investigated the feasibility of applying virtual reality (VR) and augmented reality (AR) technologies in manufacturing and material testing training programs to increase academic achievement, engagement, attitudes and self-efficacy toward engineering with the ultimate goal of accelerating progress toward degree, increasing retention and fostering more effective use of expensive lab-equipment. We hypothesized that providing students with VR and AR enabled personalized manufacturing labs would be a promising way to increase students' engagement and

better prepare them for future careers as well as broaden students' outlook on innovation. We identified the lack of VR/AR training content authoring software to be a key obstacle preventing the education sector from exploring the full potential of VR and AR for training. To address this challenge, we set out to investigate the feasibility of an authoring software that enables non-programmers to create high quality VR/AR based content. We successfully developed a software prototype that is capable of representing both equipment as well as physics simulation results as modular virtual objects that can be combined to author scenes without programming.

To validate the prototype and test our hypotheses, we conducted pilot testing at Texas A&M University (TAMU). The total number of students with complete data analyzed for this study was 118. A two-group, randomized control group experiment was conducted. Individuals who agreed to participate were randomly assigned to one of two conditions upon arriving at the study location. 57 participants were assigned to the control condition and 61 to the experimental condition. Those in the control condition performed an in-lab activity on the Instron Machine for Activity 1 and used virtual reality (VR) for Activity 2. Those in the experimental condition used the VR in Activity 1 and the in-class Instron Machine for Activity 2. Both groups were administered the tests and surveys at the same time and in the same order. No significant differences were found for pre-existing factors including pretest and demographic factors. Five measures were administered to participants including subject matter measures and survey measures relating to attitudes, self-efficacy, engagement, and background demographics. In addition to measures that students completed, action and sensor data were gathered the virtual reality session. Results of the pilot test data suggest an overall positive outcome for students participating in the VR activity first (experimental condition) compared to the Instron machine activity first (control condition). Students who participated in the VR activity first scored significantly higher on multiple choice subject matter midtest scores. Analysis of open-ended subject matter responses indicated significantly higher gains for participants in the VR activity compared to the Instron activity. Additionally, students who participated in the VR activity first reported significantly higher engagement than those who participated in the Instron machine first. Subgroup analysis indicated a different performance pattern for female participants. Specifically, when pretest to midtest gain scores were analyzed, females in the experimental group gained twice as much over males after the VR activity. Action data also revealed that female participants spent more time in the tutorial. Due to the small and uneven number of participants in the gender subgroup, caution should be taken with interpretation of results. However, these results may suggest important

performance differences between gender warranting further examination in future studies.

In summary, the pilot test results are encouraging because they show the potential of VR based lab training to engage students as well as lead to measurable learning outcomes. When viewed in conjunction with cost-savings compared to physical equipment, VR/AR based training appears highly promising as a pathway to bring advanced manufacturing training to underrepresented and underserved communities. We envision that in future, simulation based virtual object models may have widespread training applications for in-situ visualizations of phenomena that are invisible to the naked-eye.

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Modified by: Rajesh K Jha

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